Materials Thermodynamics and Kinetics

Course Structure

- Course Name: Materials Thermodynamics and Kinetics
- Course Code: MT-26002
- Course Type: Program Core Course (PCC)

✓ Lecture (L): 3
✓ Tutorial (T): 1
✓ Self-Study (S): 1
✓ Total Credits (C): 4

4 hours each week

Evaluation Strategy

- MSE (Mid-Sem): 30
- TA: 20
- ESE (End-Sem): 50

- TA: Assignments, tests, quizzes, and presentations
- Minimum 2 TAs each before and after Mid-Sem

Syllabus

- Unit 1: Introduction to Thermodynamics and 1st Law
- Unit 2: 2nd and 3rd Laws of Thermodynamics and Entropy
- Unit 3: Activity, Equilibrium Constant, and Free Energy
- Unit 4: Thermodynamics of Solutions
- Unit 5: Electrochemical Cells
- Unit 6: Thermodynamics of Defects

Course Content

- Unit 1

  Thermodynamics systems, Classification, thermodynamic variables, State functions, Process variables, Extensive and intensive properties, Energy and first law of thermodynamics, Heat capacity, Enthalpy, Heat of reaction, Hess's law, Kirchhoff's equation, Adiabatic flame temperature of the reactions, Thermochemistry.

- Unit 2

  Second law of thermodynamics, Entropy, Effect of temperature on entropy, Statistical nature of entropy, Combined statements of first and second law of thermodynamics, Gibbs' free energy, Helmholtz's free energy, Maxwell's equations, Gibbs-Helmholtz equation, Clausius-Clapeyron's equation, and its application to phase changes, Free energy as criterion for equilibrium and its applications to metallurgical reactions, Third law of thermodynamics.

- Unit 3

  Activity, Equilibrium constant, Le –Chatelier's principle, Chemical potential, Law of mass action, Effect of temperature and pressure on equilibrium constant, Vant Hoff's isotherm, Sigma function, Free energy-temperature diagrams, oxygen potential and oxygen dissociation pressure, Gibbs phase rule and its applications, Free energy composition diagram, Ellingham diagrams.

Course Content (*Conti*...)

- Unit 4

  Solutions, Partial molar quantities, Ideal solutions, Raoult's law, Non-ideal solutions, Gibbs-Duhem equation, Free energy of formation of solution, Regular solutions, application to phase equilibria, excess thermodynamic quantities

- Unit 5

  Electrochemical cell, Determination of thermodynamic quantities using reversible electrochemical cell, EMF cell, electrode potential, Electrode potential-pH diagrams and their applications

- Unit 6

  Thermodynamics of crystalline defects: surfaces and interfaces of solids, vacancies, and interstitials in solid metals, Reaction kinetics: Arrhenius equation, order of reactions

Text and Reference Books

- Text Books:
  1. D.R. Gaskell, Introduction to Thermodynamics of Materials, III Edition, McGraw-Hill Book Co. Inc.
  2. Ahindra Ghosh, Textbook of Materials & Metallurgical Thermodynamics, Prentice Hall India.
  3. S.K. Bose and S.K. Roy, Principles of Metallurgical Thermodynamics, 1st Edition, Universities Press, Hyderabad.
- Reference Books:
  1. L.S. Darken and R.W. Gurry, Physical Chemistry of Metals, McGraw- Hill, 1958.
  2. R.H. Parker, An Introduction to Chemical Metallurgy: Pergamon Press, Inc.
  3. G.S. Upadhya and R.K. Dube, Problems in Metallurgical Thermodynamics and Kinetics, Pergamon Press, Inc., 1977.

Introduction

Thermodynamics

- Physics
- Mechanical
- Metallurgy

“Therme”
Thermo

Heat
Thermal Energy

“Dynamikos”
Dynamics

Power
Motion

Thermodynamics

Motion or power of thermal energy or heat

Introduction

- Heat transfers, work done, or conversion of different forms of energy one into another
- Beyond engine – Chemical and Structural Stability
- Stability of an alloy or a crystal structure under certain conditions
- Thermodynamics of materials

Real-life Example

- **Diamond vs Graphite**
- Diamond is thermodynamically unstable at room temperature
- Graphite is the stable form of carbon at RT
- So, what is happening here?
- Thermodynamics vs Kinetics
- Meta/unstable => Stable form – Energetically favorable (Thermodynamics)
- High Activation Energy Barrier – transformation will take a long time (Kinetics)

Thermodynamics vs. Kinetics

Thermodynamics	**Kinetics**
Will a reaction or phase transformation happen?	***How fast*** will it happen, and by what path?
Initial and final states (Independent of time)	The path taken and the dimension of **Time (t)**
Driving force (ΔG – Gibbs Free Energy)	Activation Energy (Q) and Diffusion coefficients

Importance of Thermodynamics

- Possibility: Nature's ultimate filter in determining the feasibility of a chemical reaction or phase transformation under specific conditions
- Structural Stability: Stability of crystal structures, phases, or alloys at a given temperature and pressure – stable, metastable, or unstable
- Driving Force: Difference in energy (ΔG) for atoms to move, diffuse, and rearrange during processing
- Environmental Prediction: Interactions of materials with their surroundings to predicting the purity – pure, oxidize, or corrode
- Time and Resources: Predicting alloy behaviors by computational modeling to avoid performing multiple costly, blind laboratory experiments

Applications of Thermodynamics

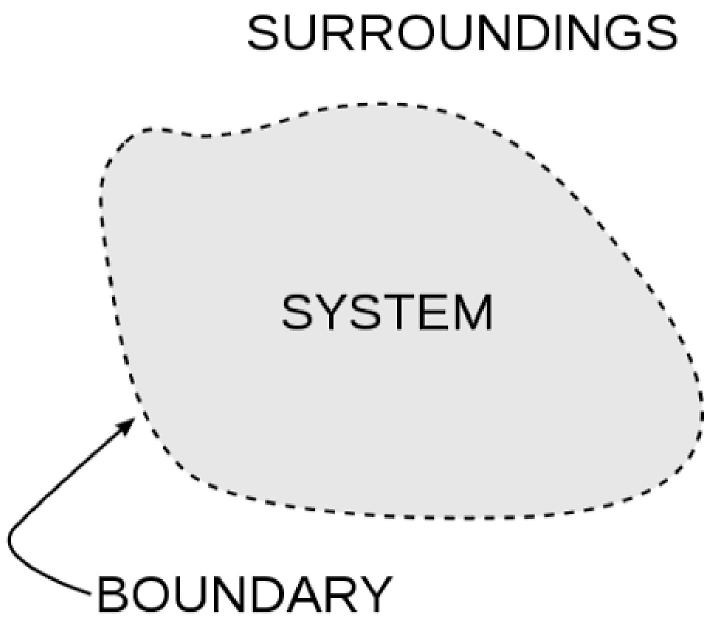
- Alloy Design and Phase Diagrams
  - ✓ Predicting Solubility: Addition of an alloying element to form a solid solution
  - ✓ Constructing Phase Diagrams: To construct different phase diagrams using thermodynamic equations
- Heat Treatment and Microstructure Engineering
  - ✓ Precipitation Hardening: Provides an idea about the temperature ranges for forming secondary strengthening particles or precipitates
  - ✓ Controlling Solidification: Shrinkage of a liquid metal, segregation, and solidification inside a mold, to eliminate defects like casting porosity
- Thin Films and Semiconductor Processing
  - ✓ Chemical Vapor Deposition (CVD)
  - ✓ Doping Control to alter the electrical properties of semiconductor devices
- Corrosion Prevention and Surface Engineering
  - ✓ Designing Stainless Steels to Prevent Corrosion by adding an appropriate amount of Chromium (Cr) to form a passive, self-healing Cr-oxide film
  - ✓ Pourbaix Diagrams to map the electrochemical thermodynamic for determining whether a metal structure (like an underground pipeline) will safely passivate, corrode, or require cathodic protection

Overview

- It determines the possibility or feasibility of a reaction or phase transformation
- If thermodynamics can't provide proper information about the transformation, it will be difficult to occur
- It provides a comprehensive understanding of the temperature range to set the furnace for any heat-treatments
- It is essential for the extractive and foundry metallurgist for melting
- To design an alloy, thermodynamics is the critical part

System and Surroundings

- System: Any portion of the Universe selected for the consideration
- Surroundings: Rest of the Universe



Types of Systems

Systems can be categorized depending on various parameters.

- Nature of interactions
- Number of components
- Reactiveness
- Phase

Types of Systems

- Nature of interactions
  - Open: Systems can exchange both energy and mass with the surroundings
  - Closed: Systems can exchange only energy with the surroundings; no mass transfer or constant mass
  - Isolated: Both energy and mass remain constant as the transfer is not possible
- Number of components
  - Unary Component: The system has a single component
  - Multi-Component System: The system may have multiple components
- Reactiveness
  - Reactive or Non-reactive: Whether the system is chemically reactive or not
- Phase
  - Homogeneous: The system has one phase
  - Heterogeneous: The system consists of multiple phases

Boundaries or Walls

- Adiabatic: No thermal energy can pass through
- Diathermal: Thermal energy can pass through
- Permeable: Mass transfer is possible
- Impermeable: Mass transfer is not possible
- Semipermeable: some of the components can pass through

Processes

- Cyclic: A sequence of processes that ***return*** to their initial position
- Adiabatic: Process where net ***heat change*** is zero
- Isothermal: Process where net ***temperature change*** is zero
- Isobaric: Process where net ***pressure change*** is zero
- Isochoric: Process where net ***volume change*** is zero

Thermodynamic Variables

- 2 types of thermodynamic state variables (or properties)
  1. Extensive variables: These properties depend on the size or mass of the system. Example: volume
  2. Intensive Variables: These properties are independent of the size and mass of the system. Example: temperature and pressure

- Extensive variables => Intensive variables (variables expressed as per unit volume or per unit mass); example: specific volume (independent of the mass of the system)

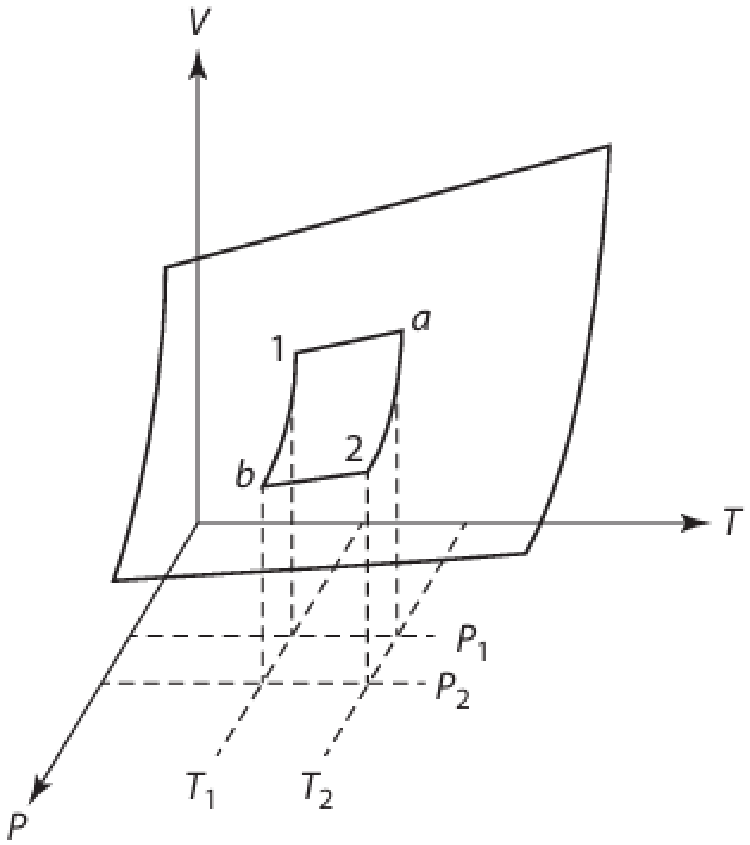
Thermodynamic Equilibrium

- **Thermal Equilibrium:** Uniform temperature throughout the system
- **Mechanical Equilibrium:** Uniform pressure throughout the system
- **Chemical Equilibrium:** Uniform chemical potential throughout the system

- **Thermodynamic Equilibrium:** If a system is in both thermal and mechanical equilibrium, then it is known as in thermodynamic equilibrium

Concept of State

- The most important concept in thermodynamics is the thermodynamic state
- Microscopic state – knowledge of masses, velocities, positions, and all modes of motion of all the constituent particles present in a system – this will determine all the measurable thermodynamic variables in the system, like energy, temperature, pressure, etc.
- It is difficult to get all this detailed knowledge in microscopic scale
- Classical thermodynamics defines the macroscopic state of a system - when all of the thermodynamic variables are fixed, then the macroscopic state of the system is fixed and is said to be in equilibrium



1 mole of pure gas
Volume = V
V = V(P, T)

Process changes from state 1 to state 2

Derive volume change from 1 => 2

- Boyle's Law
- Charles' Law

- Isothermal Compressibility (β_T)
- Coefficient of Thermal Expansion (α)

- Derive β_T with dimension of P^{-1}
- Derive α with dimension of T^{-1}

Laws of Thermodynamics

- 1^{st} Law of Thermodynamics: Energy can neither be created nor destroyed; it may only be converted from one form to another
- 2^{nd} Law of Thermodynamics: In any spontaneous or natural process, the total entropy of an isolated system always increases over time
- 3^{rd} Law of Thermodynamics: As the temperature of a system approaches absolute zero (0 K or -273.15 °C), the entropy of a pure, perfectly crystalline substance approaches zero

- Zeroth Law of Thermodynamics: If two systems or bodies are each in thermal equilibrium with a third system or body, then they are in thermal equilibrium with each other

1^{st} Law of Thermodynamics

- **Statement**
  - Energy can neither be created nor destroyed; it may only be converted from one form to another
  - The total energy remains constant
  - If a certain quantity of one form of energy disappears, an exactly equivalent amount of energy of another form appears

Heat and Work

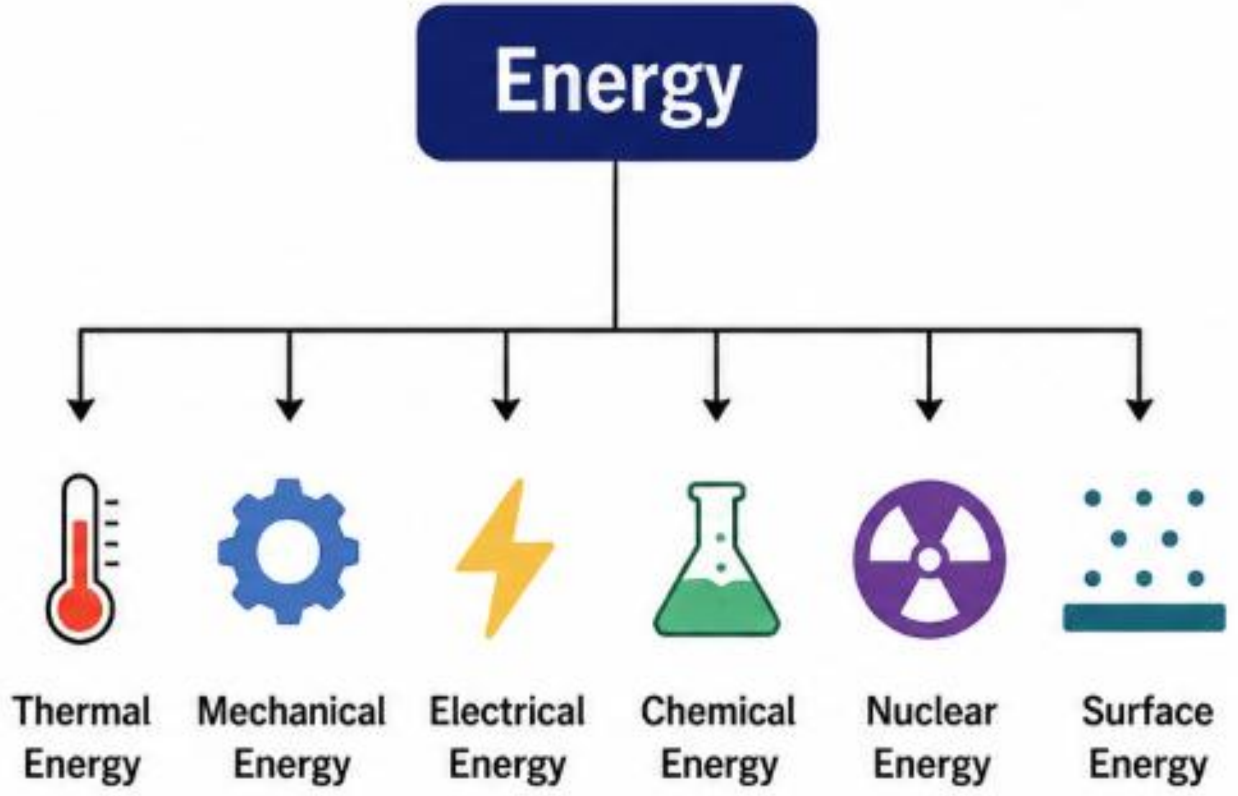
- ***Heat (q):*** Disorganized energy transfer driven solely by physical temperature gradients.
- ***Work (w):*** Organized, directional energy transfer through macroscopic displacements.
- ***q*** produced during the work and ***w*** performed during the same work
- Roughly proportional

Sign Conventions

- **Positive (+):**
  - ✓ Energy added to the system
  - ✓ Heat is absorbed ($q > 0$, endothermic)
  - ✓ Work is done *on* the system
  - ✓ Work done on the system by the surroundings (e.g., compressing a gas)
- **Negative (-):**
  - ✓ Energy lost from the system
  - ✓ Heat is released ($q < 0$, exothermic)
  - ✓ Work is done *by* the system
  - ✓ Work done by the system on the surroundings (e.g., a gas expanding)

Energy

- Capacity to do some work

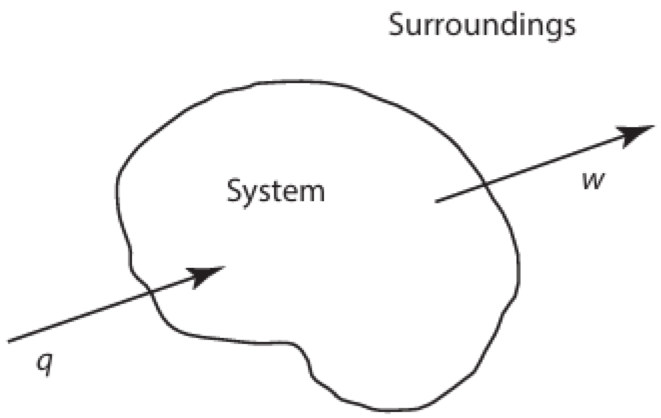


Internal Energy

- The change of a body inside an adiabatic enclosure from a given initial state to a given final state involves the same amount of work by whatever means the process is carried out
- Every system has a certain amount of energy within itself
- ***INTERNAL ENERGY (U)***
  - It represents the total microscopic kinetic and potential configurations of the metallic system, not at the macroscopic scale
  - State function of a system
  - Extensive property
  - Total energy = Potential Energy + Kinetic Energy + Internal Energy

1^{st} Law of Thermodynamics

Consider a body, initially in the state A, which performs work, ***w***, and absorbs energy, ***q***, via a thermal gradient, and, consequently, moves to the state B



The absorption of ***q*** *increases* the internal energy of the body by the amount ***q***

The performance of work ***w*** by the body *decreases* its internal energy by the amount ***w***

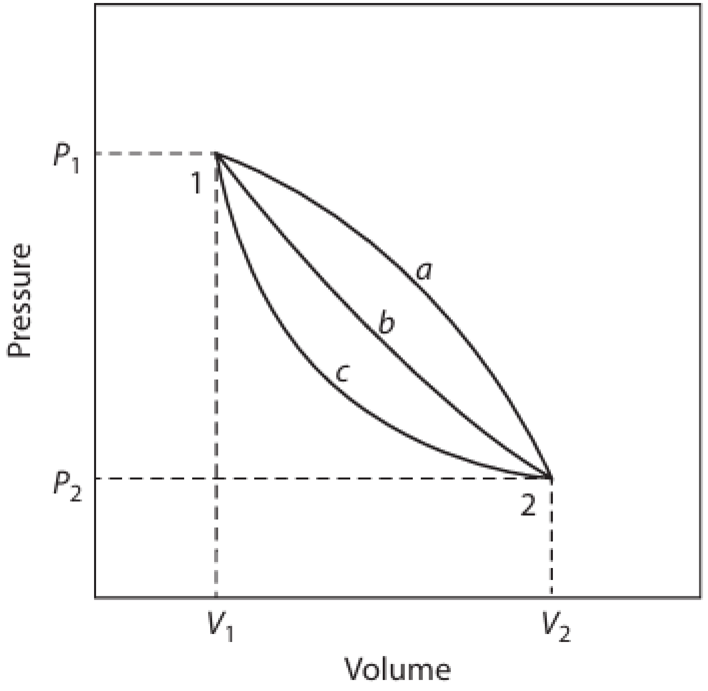
The total change in the internal energy of the body (ΔU)

$$\boldsymbol{\Delta U = U_B - U_A = q - w}$$

This is the 1^{st} Law of Thermodynamics

- For the infinitesimal change of state, the previous equation can be written as a differential
- $dU = \delta q - \delta w$

U => A Cyclic Process – State Property



$$\Delta U = \int_1^2 dU + \int_2^1 dU = (U_2 - U_1) + (U_1 - U_2) = 0$$

$$\oint dU = 0$$

For a cyclic process, where the system returns to its initial state

Therefore, it is a **state property**

U => Extensive Thermodynamic Variable

- Extensive variable
- If we consider 2 independent variables, (V and T)
- $U = U(V, T)$
- Complete differential of U in terms of the partial derivatives

Isochoric Process

- Constant volume
- The system does no work
- $\delta w = \int PdV = 0$
- δq_v is the constant-volume heat capacity

Isobaric Process

- If the pressure is maintained constant during a process that takes the system from state 1 to state 2, then the work done by the closed system is given as

$$w = P(V_2 - V_1)$$

- ***ENTHALPY (H):*** $H = U + PV$
- $dH = dU + PdV = VdP$
- At constant pressure, dP = 0 => $dH = dU + PdV$
- $\Delta H = H_2 - H_1 = q_P$
- Therefore, the enthalpy change during a constant-pressure process is equal to the thermal energy admitted to or withdrawn from the system during the process

Hess' Law

- Enthalpy is a state function – path independent
- The net enthalpy change of any process is entirely independent of the route or steps taken
- Consider an oxidation process of BCC Fe to Hematite (Fe_2O_3)

$$2Fe(\alpha) + \frac{3}{2}O_2 \rightarrow Fe_2O_3$$

Hess' Law

- Breakdown: $3Fe(\alpha) + 2O_2 \rightarrow Fe_3O_4$ ΔH_1

  $2Fe_3O_4 + \frac{1}{2}O_2 \rightarrow 3Fe_2O_3$ ΔH_2
- 2/3rd of 1st reaction and 1/3rd of 2nd reaction
- $\Delta H = \frac{2}{3}\Delta H_1 + \frac{1}{3}\Delta H_2$
- $\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3 + \cdots + \Delta H_n$
- **This is Hess' Law**
- Therefore, if a process occurs from 1->2 and if it follows 1->3->4->2, then the enthalpy change in 1->2 will be equal to the sum of the enthalpies of 1->3, 3->4, and 4->2

If I put both spoons into hot tea, which one will become hot first?

Metal conducts better

If I heat 1 kg of water and 1 kg of steel with the same heater for the same amount of time, what will happen?

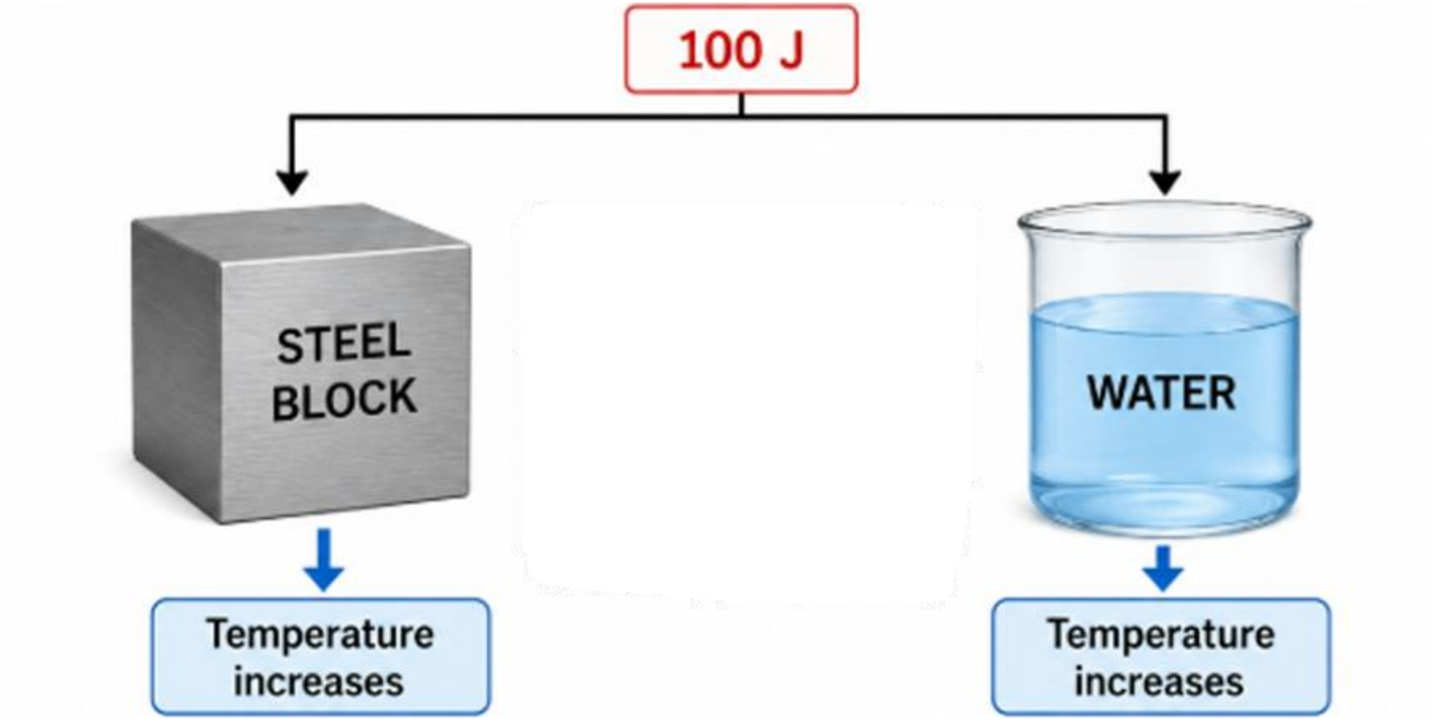
- Will both reach the same temperature?
- If not, then which one will become hot first?

Heat Capacity

Different materials require different amounts of heat to raise their temperature

This is called ***HEAT CAPACITY***

Heat Capacity



- Same amount of heat supplied
- But different temperatures increase
- For steel, ΔT = 20 °C
- For water, ΔT = 2 °C
- Due to the Heat Capacity of both materials
- In this case, water stores more heat

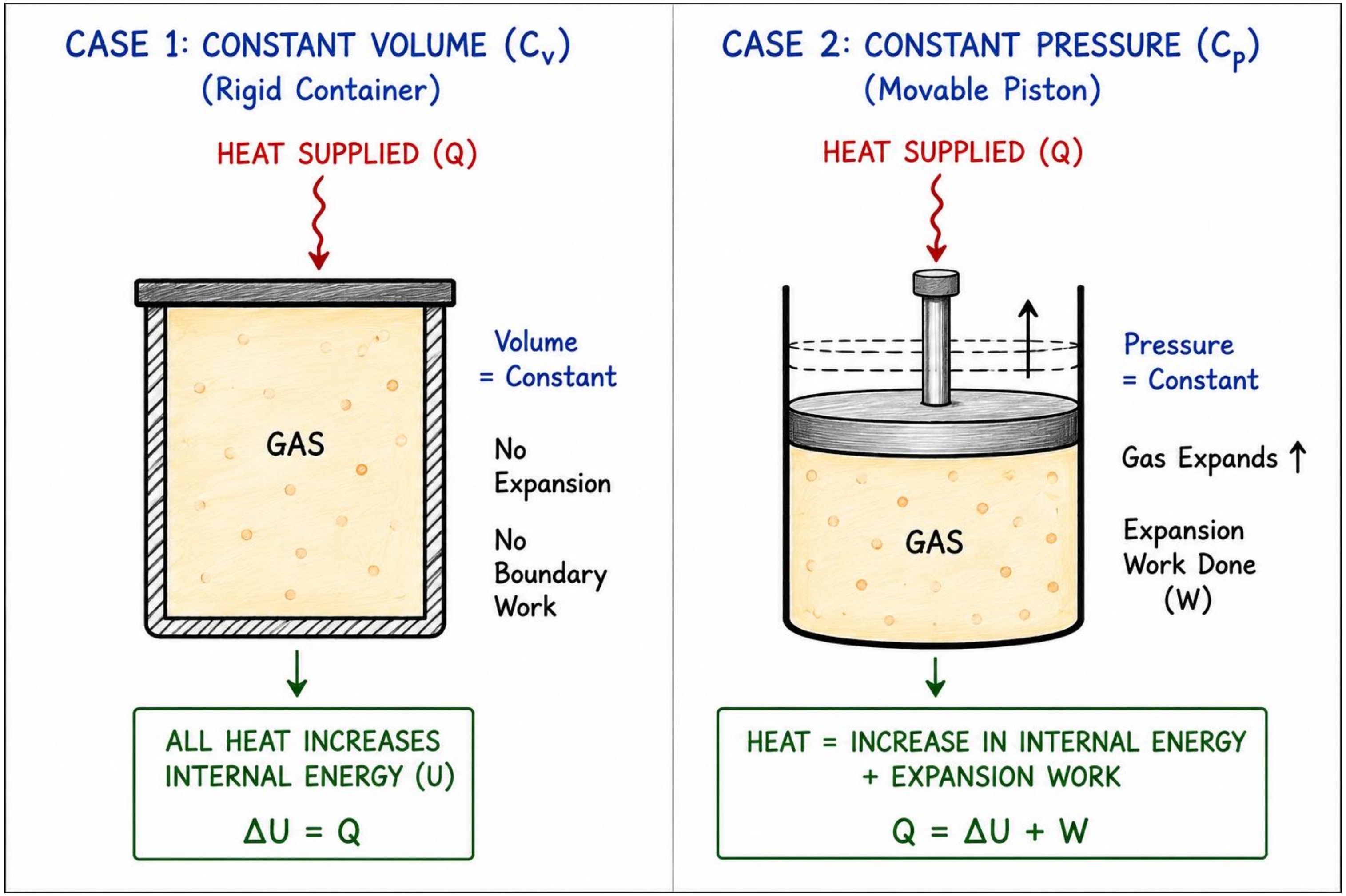
Heat Capacity

- Heat Capacity (C) is the quantity of heat required to raise the temperature of a substance by 1 °C

$$C = \frac{q}{\Delta T}$$

- For a very small change in temperature

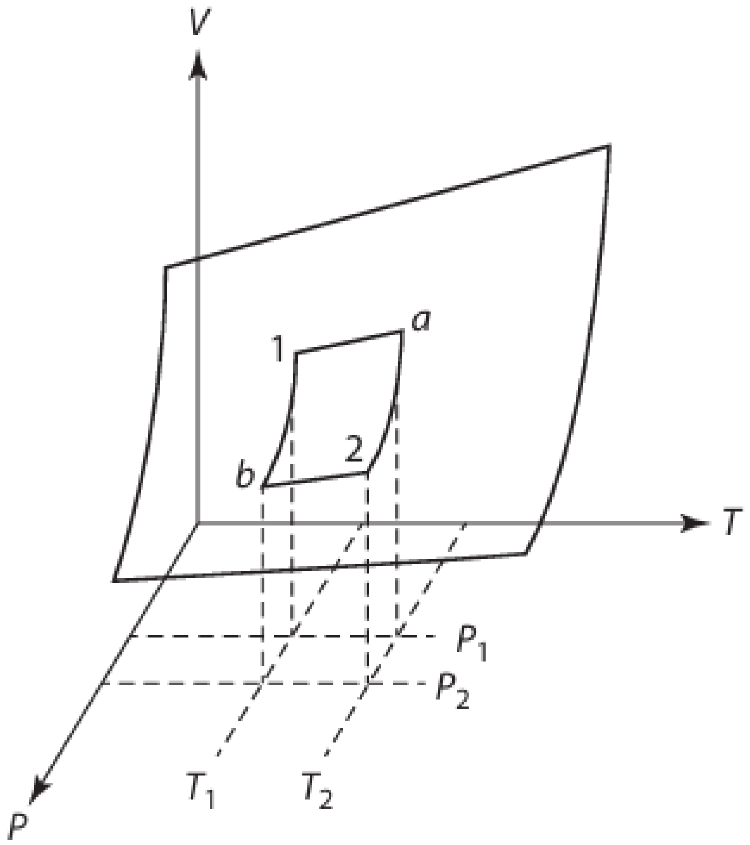
$$C = \frac{\delta q}{dT}$$


CASE 1: CONSTANT VOLUME (C_v)
(Rigid Container)
HEAT SUPPLIED (Q)
GAS
Volume
= Constant
No
Expansion
No
Boundary
Work
ALL HEAT INCREASES
INTERNAL ENERGY (U)
ΔU = Q
CASE 2: CONSTANT PRESSURE (C_p)
(Movable Piston)
HEAT SUPPLIED (Q)
GAS
Pressure
= Constant
Gas Expands ↑
Expansion
Work Done
(W)
HEAT = INCREASE IN INTERNAL ENERGY
+ EXPANSION WORK
Q = ΔU + W

Heat Capacity at Constant Volume and Pressure

- V = constant; dV = 0
- So, work done (w) = 0 (w = PdV)
- $(dU)_v = \delta q_v$
- C_v = Heat Capacity at Constant Volume
- C_p = Heat Capacity at Constant Pressure
- $C_p - C_v = R$

Reversible Adiabatic Processes



- Process changes from state 1 to state 2
- During a reversible process in which the state of a gas is changed, the gas never leaves the equilibrium surface, as shown in the figure
- As a result, during a reversible process, the gas passes through a continuum of equilibrium states, and the work w is given by the integral of PdV within the limit of V_1 to V_2
- Now, in an adiabatic process, q = 0

- **Proof** $PV^{\gamma} = Constant$

Thermochemistry

- The study of heat effects accompanying chemical reactions, the formation of solutions and changes in the state of matter such as melting or vaporization, and other physico-chemical processes.
- Applications
  - Heat of reaction
  - Heat of formation
  - Heat of combustion
  - Heat of solution

Heat of X ....

- Heat of reaction – the heat evolved or absorbed (or change in enthalpy) when the reactants react completely to produce products
- Heat of formation – change in enthalpy when one mole of the compound is formed from its constituent elements
- Heat of combustion – enthalpy change when one mole of the substance is completely burnt in O_2
- Heat of solution – the enthalpy change when one substance dissolves in another

Standard Heat of Formation

- Heat of formation of a compound in the constituent elements' standard states
- Most stable form
- Commonly at 1 atmospheric pressure (for solid/liquid/gas)
- At 298 K (25 °C)
- $\Delta H^{\circ}_{298,X}$
- Traditionally, standard heat of formation for an element is considered zero (0) at 25 °C

Example

- Ni + ½ O_2 = NiO
- $\Delta H^{\circ}_{298,X} = \Delta H^{\circ}_{298,NiO} - \Delta H^{\circ}_{298,Ni} - \frac{1}{2}\Delta H^{\circ}_{298,O2}$
- $\Delta H^{\circ}_{298,X} = ?$

Problem

- Calculate the standard heat of reaction at 25 °C (298 K) and 1 atm pressure of $3FeO = 2Al = \ Al_2O_3 + 3Fe$

in terms of per mole of Al_2O_3 formed, per mole of Fe formed, per mole of FeO reacted, per mole of Al reacted, and per g of Fe formed.

Given: $\Delta H^{\circ}_{298,FeO} = -63.3\,\frac{kcal}{mole}\left(-264.84\,\frac{kJ}{mol}\right)$

$$\Delta H^{\circ}_{298,Al_2O_3} = -400.0\,\frac{kcal}{mole}\left(-1673.6\,\frac{kJ}{mol}\right)$$

Atomic weight of Fe = 56

Problem

- Calculate ΔH°_{298} for the following reaction: $2CaO + SiO_2 = Ca_2SiO_4$

Given: $\Delta H^{\circ}_{298,f}$ for CaO, SiO2, and Ca2SiO4 from their elements at 298 K are -634, -910.9, and -2305.3 kJ/mole, respectively.

Kirchhoff's Law

- If a system undergoes a change from one state to another state, both the internal energy and heat occur would alter
- The relationship of $\Delta H°$ of a reaction with temperature and other quantities is known as Kirchhoff's law
- Another law based on the state property nature of Enthalpy
- Derive Kirchhoff's law in terms of specific heat capacity and enthalpy change
- 2 different ways (simple and standard states)

Maier-Kelley Empirical Formulation

- C_p can't be considered as a constant in real-world engineering design
- For high-temperature metallurgical calculations, C_p is modeled as a non-linear power series:

$$C_p(T) = a + bT + cT^{-2}$$

Adiabatic Flame Temperature

- If a fuel is assumed to burn under adiabatic conditions (i.e., no heat loss to surroundings), the resulting temperature of the products of combustion is known as the ***Adiabatic Flame Temperature*** (T_a)

- $Heat\ required\ to\ raise\ the\ temperature\ of\ products\ from\ 298\ K\ to\ Ta\ = Sensible\ heats\ of\ reactants\ at\ 298\ K - Heat\ released\ due\ to\ combustion\ at\ 298\ K$

- Sensible heats of reactants at 298 K are zero (0) by definition of sensible heat

Problem

- Calculate the adiabatic flame temperature when C_2H_4 gas is ignited at 298 K with a stoichiometric amount of O_2.

Given: 1. Heat of combustion (ΔH) of C_2H_4 at 298 K is -1411.3 kJ/mole

2. C_p of gases Co_2, H_2O, and O_2 are ($26.76 + 42.26*10^{-3}$ T), ($30.2 + 9.93*10^{-3}$ T), and ($25.5 + 13.61*10^{-3}$ T)